

SIVA ANJAN KUMAR THOTA

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SUMMARY

Aspiring Data Scientist and Machine Learning Engineer pursuing an M.Tech in Data Science at SRM University-AP. Experienced in developing deep learning, multimodal AI, and machine learning solutions using PyTorch, Transformers, NLP, and Computer Vision. Built research-driven models for multimodal disaster understanding, protein family classification, and phishing URL detection with a strong focus on model optimization and real-world AI applications.

TECHNICAL SKILLS

Programming: Python, SQL, Java (Basic)
Machine Learning: PyTorch, Scikit-learn, Transformers, CNNs, NLP, Feature Engineering
Data Science: Pandas, NumPy, Matplotlib, Power BI
Web & Tools: Flask, MySQL, Git, Jupyter Notebook, VS Code

PROJECTS

Reliability-Aware Multimodal Disaster Understanding Using Uncertainty-Guided Cross-Modal Fusion

PyTorch, DeBERTa-v3, CLIP ViT-B/32, GraphSAGE, Hugging Face Transformers, OpenCLIP

- Developed a multimodal disaster understanding framework integrating DeBERTa-v3, CLIP ViT-B/32, GraphSAGE and uncertainty-aware feature fusion.
- Designed Evidential Reliability Estimation (E-REM), Uncertainty-Aware Semantic Gating (UASG), and Uncertainty-Guided Cross-Modal Attention (UG-CMA) with InfoNCE contrastive learning.
- Trained and evaluated the model on 18,082 CrisisMMD v2.0 image-text pairs (14,465 train / 1,808 validation / 1,809 test).
- Achieved 0.989 weighted F1 for disaster event classification and 0.858 weighted F1 for informative content classification.
- Presented experimental findings during M.Tech thesis reviews and documented the complete training pipeline.

BioProteinFormer – Few-Shot Protein Family Classification

PyTorch, Dilated CNN, SE Attention, Multi-Head Attention, ArcFace

- Developed a hybrid protein sequence encoder integrating Dilated CNN, SE Attention, Multi-Head Self-Attention, and ArcFace for few-shot protein family classification.
- Designed a deep learning architecture to learn discriminative protein representations from amino acid sequences.
- Trained and evaluated the model on 6,000 RCSB PDB protein sequences across 12 protein families.
- Achieved 89.40% classification accuracy, outperforming 10 published baseline models by up to 11 percentage points.
- Generated 256-dimensional ArcFace embeddings achieving 79.9% accuracy in 5-way few-shot learning (K=10) and documented the complete experimental pipeline.

Phishing URL Detection Using Machine Learning

Python, Scikit-learn, Random Forest, SVM, Logistic Regression

- Developed a machine learning pipeline for phishing URL detection using Random Forest, SVM, and Logistic Regression classifiers.
- Engineered 34 handcrafted URL-based features including SSL validity, entropy, subdomain depth, redirects, and lexical characteristics.
- Trained and compared multiple machine learning models using ROC-AUC and classification performance metrics.
- Improved Random Forest accuracy by approximately 12% through custom ternary feature scaling and threshold optimization to reduce false negatives.
- Built a lightweight inference pipeline for blacklist-independent phishing detection and prepared comparative performance reports with model evaluation visualizations.

EDUCATION

M.Tech in Data Science, SRM University AP CGPA: 8.39	2025 – Present
B.Tech in Artificial Intelligence & Data Science, VVIT CGPA: 7.04	2021 – 2025
Intermediate, Narayana Junior College CGPA: 8.23	2019 – 2021
Secondary Education, St. John's High School CGPA: 7.85	2019

CERTIFICATIONS

- Data Science for Engineers – NPTEL (2024)
- Microsoft Power BI – Microsoft
- Cloud Computing – NPTEL (2024)
- Salesforce Administrator – Salesforce (2024)